

TRACY, MN, USA

WMO: 722342

Lat: 44.249N

Lon: 95.607W

Elev: 1340

StdP: 14.00

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-5.8	-16.7	2.3	-8.8	-12.8	2.9	-5.2	32.8	18.6	29.2	16.4	9.0	320	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	89.8	73.5	87.0	71.8	84.1	70.4	77.1	85.9	74.9	83.5	73.0	81.0	11.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.2	134.1	82.8	72.3	125.7	80.9	70.3	117.1	78.7	41.4	85.8	39.4	83.5	37.5	80.6	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.9	24.5	21.4	-15.7	95.1	5.1	2.4	-19.4	96.9	-22.3	98.3	-25.2	99.7	-28.9	101.5		
			-16.2	79.8	4.6	2.6	-19.5	81.7	-22.3	83.2	-24.9	84.7	-28.2	86.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	17.4	19.2	33.0	45.9	59.1	68.8	72.9	69.7	63.2	50.5	35.0	21.8	
	DBStd	22.23	13.39	12.55	12.76	10.71	9.16	6.35	5.65	5.25	8.33	9.56	11.96	11.74	
	HDD50	4092	1010	863	541	197	26	0	0	0	4	112	466	873	
	HDD65	7436	1475	1283	991	575	228	33	7	15	133	459	899	1338	
	CDD50	2835	0	0	16	75	307	566	711	612	402	128	18	0	
	CDD65	702	0	0	1	2	45	149	253	162	80	10	0	0	
	CDH74	5927	0	0	4	39	485	1246	2234	1160	663	95	1	0	
	CDH80	1747	0	0	0	5	154	376	724	285	187	16	0	0	
Wind	WSAvg	9.6	11.4	10.9	10.6	11.2	10.3	8.1	7.1	6.4	8.3	9.5	11.1	10.5	
Precipitation	PrecAvg	28.8	0.7	0.7	1.5	2.9	3.5	4.4	3.4	3.5	3.5	2.3	1.4	0.9	
	PrecMax	41.0	1.9	1.5	3.6	7.9	6.6	11.0	6.5	6.3	11.6	6.1	4.5	3.3	
	PrecMin	21.3	0.1	0.0	0.1	0.4	1.1	2.2	0.6	0.7	0.7	0.3	0.0	0.1	
	PrecStd	4.9	0.6	0.4	0.9	1.6	1.7	1.9	1.5	1.6	2.6	1.7	1.3	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	56.4	72.2	78.9	90.7	91.8	93.4	90.2	89.8	82.0	70.0	51.4	
		MCWB	39.2	50.5	56.9	58.8	65.7	72.5	76.1	75.8	70.8	62.4	55.1	46.9	
	2%	DB	41.0	45.9	63.0	72.9	84.2	87.8	90.2	86.5	85.6	76.5	62.5	44.5	
		MCWB	36.5	40.6	52.9	56.6	64.0	69.8	75.8	74.4	69.7	60.3	52.3	39.3	
	5%	DB	37.3	39.8	55.8	68.0	79.8	84.4	87.2	83.0	81.5	70.9	56.9	39.0	
		MCWB	34.5	35.8	49.3	53.4	63.0	68.6	74.3	72.0	66.5	57.9	47.8	35.5	
	10%	DB	34.3	35.8	50.3	62.9	74.6	81.3	84.0	80.6	77.4	65.3	51.6	36.2	
		MCWB	32.4	33.0	45.7	51.8	59.9	67.4	72.2	69.6	65.3	54.5	45.0	33.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.2	49.9	60.1	62.0	71.2	77.1	80.9	78.9	76.1	67.5	57.9	47.6
			MCDB	45.0	56.6	67.8	72.1	83.0	86.1	88.5	87.4	84.1	76.4	66.3	53.8
2%		WB	37.0	41.1	54.2	59.1	67.9	73.8	77.8	75.9	73.0	62.4	52.8	39.1	
		MCDB	40.5	46.0	60.0	69.7	78.4	83.9	86.8	83.8	80.1	72.3	62.0	42.6	
5%		WB	34.8	36.2	50.0	56.1	65.0	71.2	75.8	73.8	70.0	59.4	48.7	36.0	
		MCDB	37.5	39.6	57.0	64.7	75.4	80.4	85.0	81.2	78.1	68.2	55.9	38.8	
10%		WB	32.6	33.0	45.3	53.1	62.6	68.9	73.7	71.7	67.2	55.7	44.5	33.8	
		MCDB	34.1	35.3	50.4	60.9	71.2	78.0	82.1	78.4	74.9	63.9	51.2	35.9	
Mean Daily Temperature Range		5% DB	MDBR	14.5	15.6	15.3	18.6	19.4	18.2	18.5	17.9	20.4	18.8	16.1	13.7
			MCDBR	16.3	19.6	24.0	27.5	26.1	22.4	20.2	19.8	23.3	24.6	22.9	16.1
	MCWBR		13.4	14.5	15.5	15.0	12.0	10.3	10.1	10.5	11.0	13.1	13.6	12.6	
	5% WB	MCDBR	15.5	18.6	22.6	23.7	21.5	19.8	19.0	17.8	20.6	21.7	22.1	14.6	
		MCWBR	12.6	14.0	15.1	15.0	11.4	11.3	10.7	10.7	11.8	13.0	13.7	11.8	
Clear-Sky Solar Irradiance	taub	0.261	0.273	0.306	0.360	0.379	0.395	0.405	0.406	0.363	0.321	0.283	0.265		
	taud	2.459	2.438	2.449	2.335	2.350	2.364	2.320	2.290	2.388	2.495	2.524	2.460		
	Ebn at Noon	280	295	297	286	283	278	273	268	273	273	269	266		
	Edh at Noon	22	27	31	38	38	38	39	39	33	26	21	20		
All-Sky Solar Radiation	RadAvg	593	926	1216	1507	1739	1938	2093	1725	1359	899	598	448		
	RadStd	62	102	119	148	127	134	73	136	99	116	49	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page